Ge Yan

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PhD candidate, Electrical Engineering, National University of Singapore, Singapore

PROFESSIONAL SUMMARY

I am a researcher specializing in mathematical modeling, optimization, computational electromagnetics, and machine learning for scientific and engineering applications. I develop and implement algorithms for wireless communication, sensing, and inverse problems, with experience in learning-based and model-driven methods for wireless and vision applications.

TECHNICAL EXPERTISE

- **Machine learning:** deep learning, computer vision, PyTorch, TensorFlow.
- **Algorithms:** optimization, numerical methods, signal processing, inverse problems.
- **Programming & tools:** Python, MATLAB, C/C++, COMSOL, CST, CVX, ANSYS, Git, Linux.
- **Wireless:** MIMO, beamforming, channel models, computational electromagnetics.

EDUCATION

- **2022/08-2026.12 (expected)** **PhD. in Electrical and Computer Engineering**
  - Institution: National University of Singapore, Singapore
  - Advisors: Prof. Rui Zhang, Prof. Xudong Chen
  - Thesis Title: Reconfigurable MIMO for Wireless Communication: Optimization and Practical Channel Knowledge Acquisition
- **2021/01-2022/06** **MSc. in Electrical and Computer Engineering**
  - Institution: National University of Singapore, Singapore
  - GPA: 4.75/5.00
- **2016/09-2020/06** **B.Eng. in Information Science & Engineering**
  - Institution: Chien-Shiung Wu College, Southeast University, Nanjing, China
  - GPA: 3.89/4.00 (Overall Ranking: 1/114)

INDUSTRY EXPERIENCE

- **2020/08-2020/12** **AI Research Intern, R&D, SenseTime, Shanghai, China**
  - Worked on image recognition and segmentation projects.
  - Investigated GAN-based video generation.

RESEARCH EXPERIENCE

- [Ongoing] **Channel Knowledge Map (CKM) Modelling from An Electromagnetic Perspective** (*2026/04-present*)
  - Develop an electromagnetic approach to CKM modeling and reconstruction, integrating physics-informed machine learning techniques for efficient CKM representation and environment-aware inference.
  - Develop physically consistent representations for wireless channels without requiring explicit geometric descriptions of the environment.

- [Ongoing] **Slow Movable Antenna (MA)-Aided Wireless Communications Based on Long-Term Channel Statistics** (*2024/04-present*)
  - Developed statistical models for spatially varying wireless channels across different timescales, enabling efficient MA system optimization using long-term rather than instantaneous channel information.
  - Developed low-complexity optimization methods that reduce the need for frequent channel acquisition and antenna repositioning, facilitating their practical deployment.
- **Low-Overhead Configuration for Intelligent Reflecting Surface (IRS)-Assisted Wireless Systems** (*2022/10-2025/09*)
  - Developed practical channel acquisition and quasi-static optimization methods for IRS-aided wireless systems, enabling cost-efficient long-term coverage enhancement.
  - Utilized channel measurements available from existing protocols without additional signalling, facilitating seamless integration of IRSs into existing networks.

SELECTED PROJECTS & IMPLEMENTATIONS

- **Movable Antenna (MA)-Aided Wireless Communications Prototype** (*2024/12-2025/09*)
  - Developed a prototype featuring mechanical antenna-positioning control and integrated it with National Instruments software-defined radio platforms.
  - Implemented and validated research algorithms under practical system conditions, including algorithm integration, debugging, and performance evaluation.
- **Intelligent Reflecting Surface (IRS) Channel Acquisition Prototype** (*2024/05-2024/10*)
  - Collaborated with a remote implementation team to integrate algorithms from my research into a prototype of IRS-aided wireless communication systems.
  - Provided algorithm implementation and technical support, assisted with system debugging, and contributed to performance evaluation and validation.

SELECTED PUBLICATIONS

1. **G. Yan**, L. Zhu, W. Ma, and R. Zhang, "Slow Movable Antenna System Design Based on Cell-Specific Long-Term Angular Power Spectrum," 2026. [Online]. Available: https://arxiv.org/abs/2605.12192. (submitted to *IEEE Trans. Wireless Commun.*)
2. **G. Yan**, L. Zhu, and R. Zhang, "Movable Antenna Aided Multiuser Communications: Antenna Position Optimization Based on Statistical Channel Information," in *IEEE Trans. Commun.*, vol. 74, pp. 1793-1810, 2026.
3. **G. Yan**, L. Zhu, and R. Zhang, "Power Measurement Enabled Channel Autocorrelation Matrix Estimation for IRS-Assisted Wireless Communication," in *IEEE Trans. Wireless Commun.*, vol. 24, no. 3, pp. 1832-1848, Mar. 2025.
4. **G. Yan**, L. Zhu, H. Sun, and R. Zhang, "Wideband Coverage Enhancement for IRS-Aided Wireless Networks Based on Power Measurement," in *IEEE Wireless Commun. Lett.*, vol. 15, pp. 670-674, 2026.

AWARDS/FELLOWSHIPS

- President's Graduate Fellowship (*2022/08-2026/07*) National University of Singapore
- National Scholarship (Top 1%, *2018/10*) Chinese Ministry of Education
- President Scholarship (Top 1%, *2017/10*) Southeast University, Nanjing, China